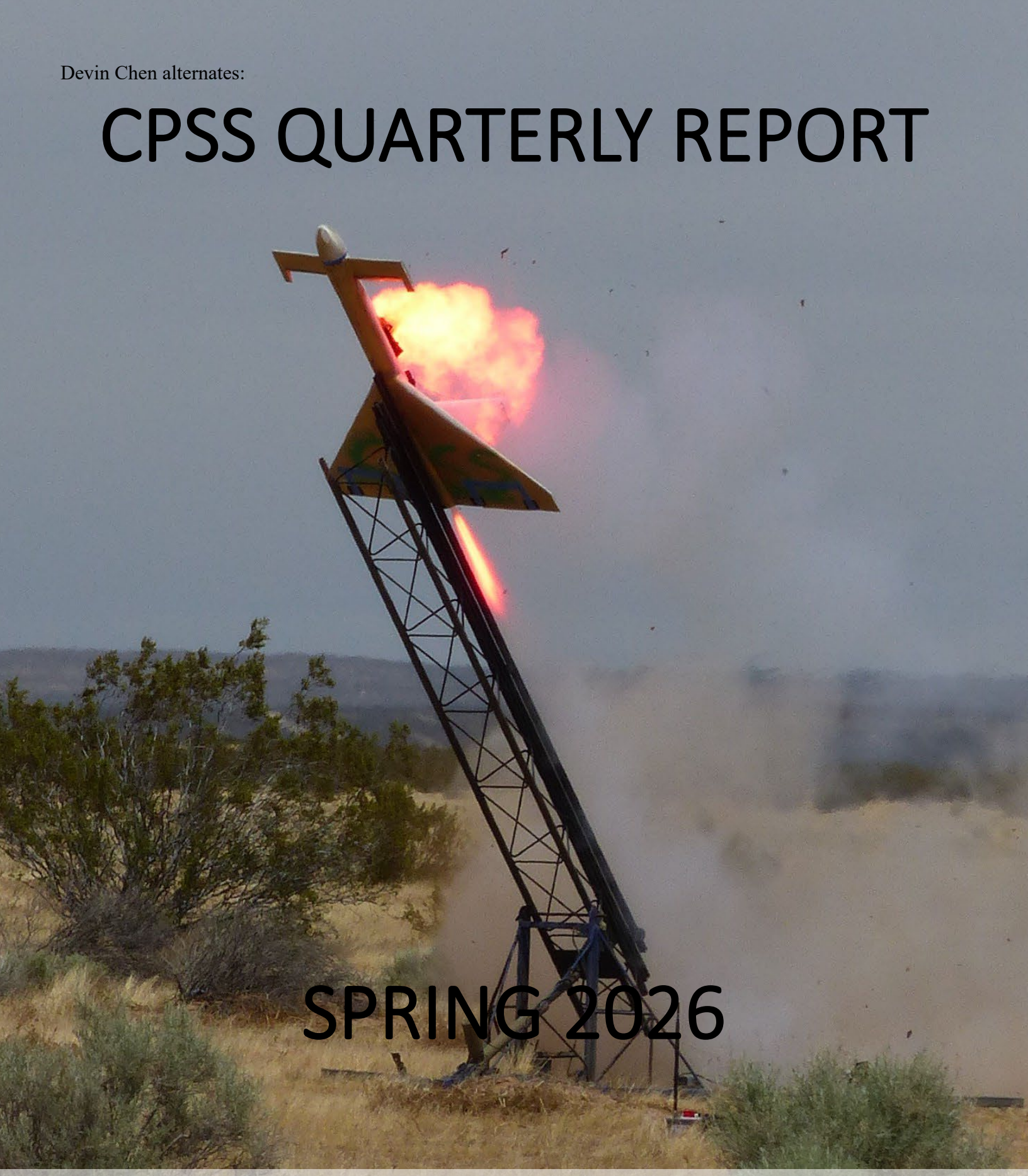
Devin Chen alternates:

CPSS QUARTERLY REPORT



SPRING 2026

Cal Poly Space Systems
California Polytechnic State University
San Luis Obispo, California, 93407

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I. MISSION

The principal objectives of CPSS in the 2025-2026 academic year are as follows:

- Demonstrate the successful operation of air brakes on a launch vehicle for apogee correction.
- Launch and recover a liquid rocket to 15,000 feet in the FAR Unlimited apogee competition.
- Launch and recover a solid motor space plane for demonstration of fly-back solid booster capability.
- Test fire a pump-fed liquid rocket engine and continue liquid engine/propulsion ground testing.
- Initiate the Tiny Rocket Program (SILL-E) for rapid design and hot fire of small-scale rockets.
- Continue the certification rocket program to get students L1 and L2 certifications.

2. OUTREACH

2.1. REDWIRE VISIT



This quarter, members of Cal Poly Space Systems had the opportunity to visit Redwire's facility in San Luis Obispo. Redwire also generously donated C motors, which will support the Booster Flyback Program and operations on the Liquids Test Stand.

Figure 1: Picture of CPSS at Redwire tour

2.2. CAL POLY INDUSTRY ADVISORY BOARD

This quarter, Cal Poly Space Systems met with the Aerospace Advisory Board to discuss the future of aerospace on campus and generate industry support. This was a great opportunity for us to present our projects for the school year and illustrate how we put Learn by Doing into practice.

2.3. INTERESTED IN GETTING INVOLVED?

If you are a potential industry partner and interested in collaborating for the 2026-2027 school year, reach out to our email: cpss@calpoly.edu. We accept monetary and material donations through official university channels. We also host industry partners for company events at club meetings, recruitment events, tours, and design reviews.

If you are a student looking to get involved, come to one of our general meetings on Mondays at 6:10 pm in 192-321 or our workdays on Saturdays at 10:10 am at the same location! Also check out our Instagram @calpolyspacesystems for more updates!

3. ENGINEERING UPDATES

3.1. LIQUID ENGINE PROGRAM

During Spring 2026, the CPSS Liquid Propulsion Team focused on improving the efficiency, proficiency, and safety of testing operations for our *Bluegrass test stand*. Through several cold flow and static hot fire tests of Stratum, the team's 300 lbf bi-propellant engine features a back pressurized fuel system; members refined test documentation and operating procedures. Development of the team's 33,000 rpm electric centrifugal turbopump

continued, including the design of a larger pump sized to meet flight vehicle requirements. Testing produced an experimental pump curve for comparison with the pump's mathematical performance model. Further development and integration will continue during the 2026–27 academic year in preparation for installation on the flight vehicle, *Archerfish*, currently scheduled to launch at Friends of Amateur Rocketry (FAR) in December 2026.

The team also established its next major objective: becoming the first collegiate rocketry team to complete the vertical takeoff and landing of a liquid propellant flight vehicle through the Collegiate Propulsive Lander Challenge. Initial work will begin during Summer 2026 with research and preliminary design focused on *thrust vector control*, *engine throttling systems*, and *regeneratively cooled engines*.



Figure 2: Stratum (Fuel Back-pressed) Hot Fire on Bluegrass Test Stand

3.2. COMPETITION PROGRAM

Archerfish, our liquid rocket, is aimed at being the first collegiate pump-fed liquid rocket while combining many previous liquid projects developed by CPSS in the last few years into a flight vehicle. The rocket will run on a bi-propellant engine (nitrous oxide and ethyl alcohol) and is set to reach an apogee of 13,000 feet, where the fuel is pumped.

So far, the composite body tubes, thrust-chamber assembly (TCA), and propellent tanks have been made. Our TCA has been successfully hydrostatic tested to withstand 800 psi and is currently undergoing cold flow testing as seen below. Our plans for next semester include finishing manufacturing, hydrostatic pressure testing our

propellent tanks, completing a successful cold flow with our TCA, testing our pump, cold flowing our full liquid system, hot firing our engine, then launching!



Figure 2: Archerfish Liquid Engine Cold Flow

3.3. NILE



Figure 3: NILE Control Room



Figure 3: Successful hot fire of SILLE Engine

NILE (Nitrous Oxide and Isopropyl Liquid Engine) Stand is a rocket engine test stand in El Paso de Robles, California. The site is part of a growing spaceport initiative focused on advancing the commercial space industry.

Designed to support engine testing up to 1,000 lbf of thrust, NILE is ideal for long-duration, low-thrust burns with a high propellant capacity, which will support several upcoming propulsion and vehicle-development projects.

Last spring, NILE became the first collegiate team to conduct a successful hot fire at the facility. Despite several technical and logistical challenges, the team completed the hot fire on a 35-lbf rocket engine, demonstrating that the stand is operational and capable of supporting more ambitious testing campaigns.

The combined efforts of NILE and local companies conducting on-site testing contributed to the spaceport receiving a $1 million contract through the NASA Ames program. This funding will support continued facility development and expand its role in commercial-space research and education.



Figure 4: NILE Control Room Monitor Station

3.4. FLY-BACK BOOSTER PROGRAM

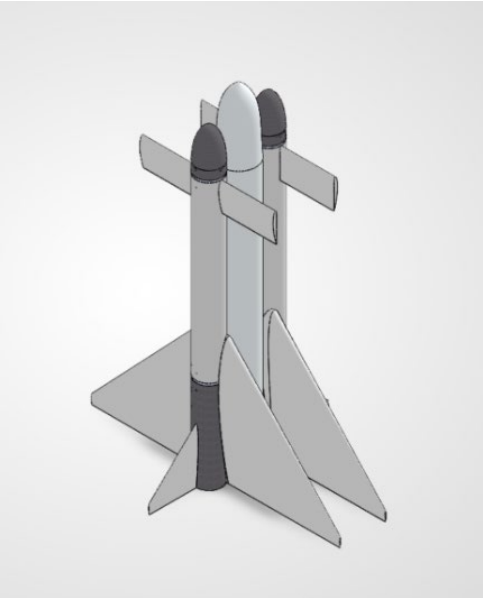


Figure 5: CAD of Booster with a centerstage rocket



Figure 6. Booster Team testing at FAR

CPSS has a long history of designing and launching Space Shuttle-inspired launch vehicles made up of a center stage mounted with two rocket-boosted, radio-controlled, glide-back boosters. These remarkable projects, first of their kind, were a tremendous success. From motor ignition to the boosters separating from the center stage to the pilots landing the boosters safely— every detail had to be done right.

Fast forwarding to now, CPSS has resurrected the idea in 2025-2026 and is seeking to launch the boosters-center stage configuration in Spring of 2028.

The academic year of 2025-2026 was spent designing the central airframe of the booster, creating the CAD, and devising the manufacturing techniques necessary for fabricating the booster. All this work conglomerated into a launch in Spring 2026; however, the team did not see the success they expected. Due to a motor misfire, the vehicle experienced catastrophic rocket failure on the rail and did not have a chance to glide to the ground.

As the team enters the new academic year, they are equipped with a tremendous amount of experience from their last launch. With this newfound launch experience, each team is seeking to build on their work from last year and optimize the booster design to be smaller, lighter, and easier to manufacture without sacrificing its flight characteristics.

3.5. AIR BRAKES PROGRAM

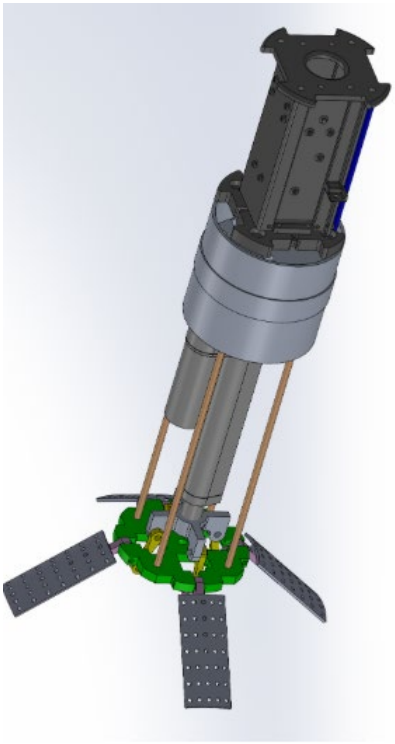


Figure 7: CAD model of Bebop's air brake mechanism

The objective of this program is to develop an active control system for our competition rockets to more accurately target an apogee and win more points at FAR's 51025UL apogee competition. The mechanism is actively controlled with a PID system to change the flap deployment angle based on kinematic data from the flight computer. Going into the spring quarter, the mechanical design of the airbrakes was complete, so the progress revolved around getting the electronics up and running. This quarter, the Airbrakes team made substantial progress toward a fully integrated and flight-ready control system. The team successfully completed the bring-up of its custom flight computer, the Telemetry Operations Module (TOM), and integrated the full suite of onboard sensors, including the inertial measurement unit (IMU), altimeter, and GPS. The team also conducted LoRa communication testing to evaluate the reliability of the system's wireless telemetry capabilities. The team also integrated its PID control implementation with the linear actuator and tested the combined system using simulated sensor data. The individual software processes were then brought together using a custom shared memory library, allowing sensor data, control calculations, telemetry, and actuation processes to communicate efficiently. Collectively, these accomplishments established a strong foundation for continued system validation and future flight testing.

3.6. TOM FLIGHT COMPUTER

Last year, CPSS engineers designed and manufactured a new edition of the flight computer titled “Telemetry Operation Manager” (TOM). Inspired by BPS Space’s AVA flight computer, TOM is an adaptable, all-in-one avionics board that can be used in a variety of future CPSS competition and research/development missions.

Just like the previous edition of the board, TOM includes necessary sensors such as altimeters, IMUs, and GPS, as well as a LoRa radio for communication with the ground station. The biggest change from the previous edition is going from multiple smaller microcontrollers to handle data processing logging, telemetry transmission, to a new Raspberry Pi CM5. This gives us much more compute power than previous editions as well as significantly reducing the barrier to entry for new members being able to do meaningful programming for avionics projects.

Over the course of last year, we assembled and validated the new version of TOM that will be performing the bulk of the heavy Avionics work for both the Bebop and Archerfish missions

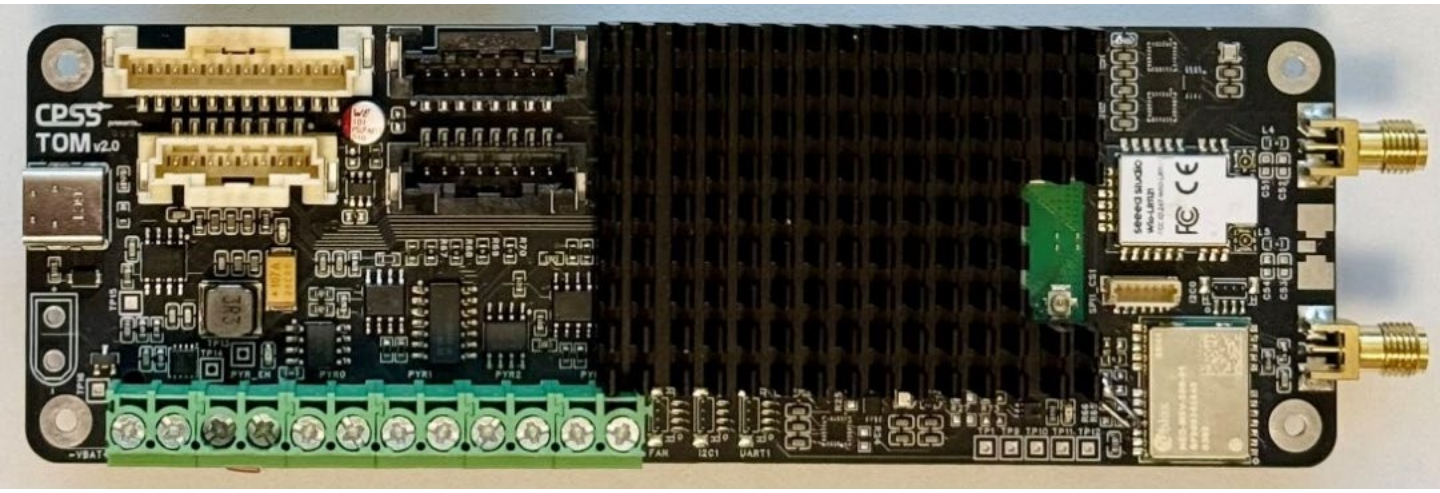


Figure 8. TOM

4. EDUCATION AND CERTIFICATION

4.1. CERTIFICATION ROCKETS

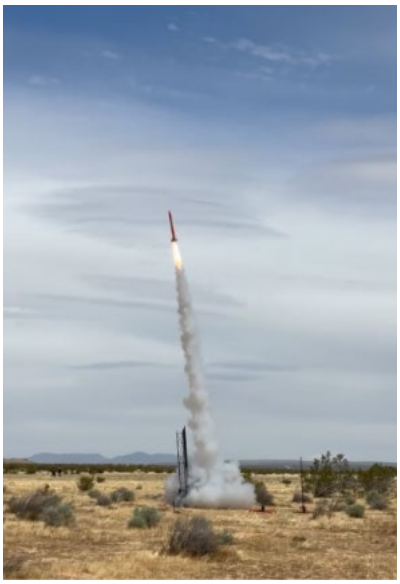


Figure 9. L2 launch at Mojave Desert Amateur Rocketry Society launch site, Spring 26



Figure 10. CPSS members with L2 at Mojave Desert Amateur Rocketry Society launch site, Spring 26

CPSS continues to build and improve the certification rocket program with many new members obtaining their L1 certifications and continuing members obtaining their L2 certifications. The program still follows the 2–4-hour workdays sessions each week until launch. Currently the program is getting more interest from students working on experiment rockets and rockets with unconventional recovery systems. There are plenty of opportunities and resources within the program for students to design and experiment with personal rockets.

Currently, the L1 kit is designed to be simple and easy to build. The design focuses primarily on the rocket having a low, slow, and steady flight with a stable assembly to ensure the best chance of having a successful certification flight. The L1 kit can be converted to an L2 kit by adding an avionics bay allowing for dual separation and dual deployment, and the L2 kit is available to launch a L2 with single deployment.

Last spring there were about 15 students who obtained their L1 certifications after constructing the updated L1 kit and one student who obtained their L2 certification after modifying a personal L1 rocket to include an avionics bay and dual deployment set up.

Finally, once more, CPSS (the certification program in particular) would like to extend a warm thank you to SEDS, as much of the progress of the certification program would not have been possible without the gracious grants from SEDS.

Interested students looking to gain their L1 or L2 Tripoli rocketry certification are encouraged to contact CPSS through our organization's email cpss@calpoly.edu.

4.2. SILL-E PROGRAM



Figure 9: SILL-E Lecture

SILLE (Student Introductory Liquids Learning Engine) is an introductory project designed to build a foundational understanding of liquid rocket engine design, alongside practical skills such as teamwork, manufacturing, testing, and best safety practices. Students participating in this project are learning about the fundamental variables that effect engine performance and are learning how to solve engine specifications and dimensions with the use of MATLAB or Python. Teams will then move on to modeling their engine in SolidWorks and give a preliminary design review (PDR) to ensure their design meets requirements. Once their PDR passes, students are advised on best machining practices and proceed to manufacturing their engine. Finally, when the team's engine is completed and approved, mentors experienced in testing will teach them about test procedures and safety. When SILLE students show proficiency in testing operations, mentors will oversee testing of their SILLE engines up to 50 lbf of thrust. One team has successfully designed, created, and tested their SILLE as seen on the figure 3 on the NILE test stand, and we have more on the way this upcoming school year.

While students new to the club are encouraged to explore all CPSS has to offer, this project is designed to improve member retention and education by lessening the barrier to entry into bipropellant engine testing. Groups have assigned mentors that oversee their project development and are available as resources to students, creating an environment to foster education and community within the club. This project aims to introduce new members to the world of liquid rocketry and encourages them to contribute positively to the growth and success of the CPSS liquids team.

4.3. CPSS EDUCATION AND EVENTS

CPSS heavily focused on onboarding and new student retention this quarter. Our expanding list of projects attracted a lot of new students. Through the introduction of our new SILLE project and our continued certification program, we hope to keep new students engaged in personal projects to develop skills in rocketry and build confidence. On our Saturday workdays, we also held introductory lectures to introduce the basics of rocketry and system design to support their practical experience with the theory behind it.

This year, we added a new social coordinator to our leadership team to encourage team growth. We hope to increase our membership and provide a fun environment to keep working on rocketry projects. Some of the events that have happened are Bowling night, Artemis II watching, BBQ party, and many more events.



Figure 10 and 11: Bowling Night and BBQ Party

5. CONTACTS

5.1. SUBJECT MATTER EXPERTS

Brian Payne	Air Brakes Project Manager	bepayne@calpoly.edu
Brian Payne	Archerfish and Bebop Flight Software Leads	bepayne@calpoly.edu
Jasmine Cheng	Archerfish Rocket Project Manager	jcheng74@calpoly.edu
Nolan McMahan	Propulsion/Ground Testing Lead	nmcmahan@calpoly.edu
Gustavo Chavez	NILE Lead	gchave28@calpoly.edu
Shiva Pulmamidi	Recoverable Booster Project Manager	spulmami@calpoly.edu
Margot Peterson	Certifications Program Manager	drsok@calpoly.edu
Lishen Lin	SILL-E Program Manager	llin34@calpoly.edu
Ellen Poulsen	SILL-E Program Managers	ecpoulse@calpoly.edu

5.2. DONATIONS AND OUTREACH

Arjan Reyes	CPSS PR Officer	areye162@calpoly.edu
Noah Shefer	CPSS Treasurer	nshefer@calpoly.edu

5.3. CAL POLY SPACE SYSTEMS

Cal Poly Space Systems	Official Organization Email	cpss@calpoly.edu
Spencer Biernat	CPSS President	sbiernat@calpoly.edu
Devin Chen	CPSS Vice President	dchen110@calpoly.edu
Metztli Singha	CPSS Secretary	msingha@calpoly.edu
Calla Woodruff Lyons	CPSS Safety Officer	cawoodru@calpoly.edu

5.4. ADVISORS

Dr. Diane DeTurris	CPSS Faculty Advisor	ddeturri@calpoly.edu